

CINEMAVE OTT

Enterprise Customer Analytics & Real-Time Data Intelligence Platform

AI-Driven Customer Analytics, Churn Prediction & Scalable Data Architecture

18

SLIDES

AI

POWERED

360°

ANALYTICS

Confidential — For Executive Review Only

Setting the Scene



Who is CineWave?

A fictional OTT streaming platform operating in a fiercely competitive digital entertainment market, facing pressure from Netflix, Amazon Prime Video, Disney+ and Apple TV+.



Why Analytics Matters

Modern OTT platforms rely on customer analytics, personalization, and real-time engagement insights to improve retention, maximise CLV, and drive sustainable growth.



Our Proposition

This case study designs an enterprise-scale analytics and data intelligence solution — from raw event streaming to AI-driven business recommendations.

Why CineWave Must Act Now

01 Subscriber Growth Plateau

Organic growth has stalled as market saturation increases

02 High Customer Churn

Subscribers cancel at alarming rates without early intervention

03 Rising Content Costs

Production budgets soar while ROI per title is declining

04 Recommendation Fatigue

Generic suggestions fail to re-engage hesitant viewers

05 Limited Personalisation

One-size experiences reduce satisfaction and time-on-platform

06 Privacy & Compliance Risk

GDPR/CCPA obligations require robust data governance

Business Impact: Reduced retention · Lower engagement · Revenue instability · User dissatisfaction

What We Set Out to Achieve



Analyse User Engagement

Uncover granular patterns in watch time, session frequency, and content preference across all segments.



Detect Churn Risks Early

Deploy predictive models to flag at-risk subscribers weeks before cancellation.



Improve Retention

Trigger personalised retention campaigns using behavioural signals and lifecycle stage.



Scalable Data Architecture

Design a modern data lake and streaming pipeline capable of processing millions of daily events.



AI Recommendation Engine

Build hybrid collaborative + content-based recommendation strategies to combat fatigue.



Real-Time Analytics

Enable live dashboards so product and business teams make data-driven decisions instantly.

Six Pillars of Customer Intelligence



BEHAVIOUR

Analytics

Watch patterns, session depth, content affinity



ENGAGEMENT

Analytics

Active minutes, click-through rates, interactions



EXPERIENCE

Analytics

CSAT, CES, NPS and friction point analysis



CHURN

Analytics

Early risk scoring, inactivity signals, downgrades



RECOMMENDATION

Analytics

Collaborative & content-based filtering models

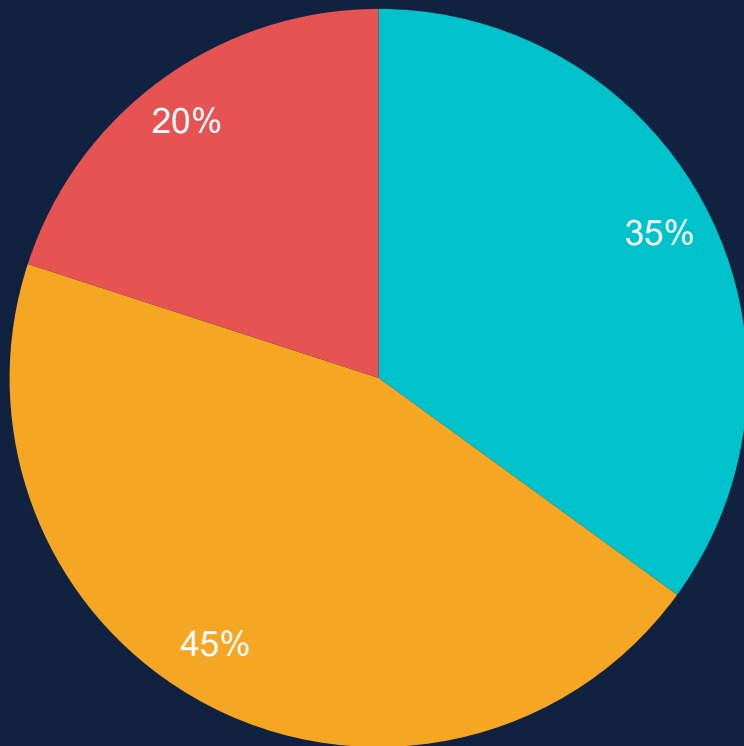


CLV

Analytics

Lifetime value modelling and tier optimisation

Know Your Viewer Types



Binge Watchers

35%

Highest engagement, lowest churn risk. Consume 3+ episodes per session. Key drivers of viral content spread.

Regular Viewers

45%

Moderate engagement, 2–4 sessions per week. Retention campaigns most effective for this group.

Casual Viewers

20%

Highest churn risk. Watch < 1 hr/week. Require targeted re-engagement and personalised nudges.

Detect, Predict & Retain

Key Churn Indicators

- Reduced watch frequency — fewer sessions per week
- Extended inactivity — no login for 14+ days
- Subscription downgrade — plan tier reduction
- Low recommendation interaction — content skipping
- Negative customer feedback — low CSAT/CES scores

AI-Driven Retention Strategy

01 Early Risk Detection

ML models score every subscriber weekly based on 30+ behavioural features.

02 Personalised Campaigns

Automated outreach triggers at risk-score thresholds — email, push, in-app.

03 Re-Engagement Nudges

Content recommendations tailored to historical preferences restart viewing habits.

Smarter Personalisation at Scale



Collaborative Filtering

User-to-User Similarity

How it works:

Identifies subscribers with similar watch histories and taste clusters. Recommends titles highly rated by analogous users.

✓ Discovers content the user didn't know they'd enjoy.



Content-Based Filtering

Metadata Intelligence

How it works:

Analyses genre, cast, director, mood, and keyword metadata. Matches content attributes to each user's preference profile.

✓ Reliable for new users with limited history.



Hybrid Engine

Best-of-Both Fusion

How it works:

Combines collaborative and content signals with engagement recency weighting and A/B tested ranking strategies.

✓ Highest precision and coverage — production-ready.

Measuring What Subscribers Feel

CSAT

Customer Satisfaction Score

4.2 / 5.0

Current Platform Score

- Post-viewing satisfaction surveys at session end
- NPS measurement across subscriber cohorts
- Genre-level CSAT breakdown to identify content gaps

CES

Customer Effort Score

3.8 / 5.0

Ease of Content Discovery

- Measures friction in search, browse, and playback
- Identifies UI flows causing drop-offs before play
- Guides UX redesign prioritisation decisions

End-to-End Pipeline Overview



Real-Time Event Streaming Layer

Watch Events

Play, pause, skip, complete — timestamped per user session

Search Activity

Query strings, zero-result searches, search-to-play conversion

Recommendation Clicks

Click-through on AI-surfaced titles and carousel interactions

Session Behaviour

Login time, idle duration, device switches, exit points

Why Kafka?

1M+

Events/sec capacity

99.9%

Uptime SLA

ms

Latency at scale

Distributed Big-Data Processing Engine

Batch Analytics

Nightly aggregation of watch history and CLV re-computation across all subscribers.

Stream Analytics

Micro-batch processing of Kafka events for near-real-time churn signal updates.

Feature Engineering

Churn model features: inactivity days, session velocity, genre entropy, platform switching.

Segmentation

K-means and hierarchical clustering to maintain dynamic viewer segment assignments.

Real-Time Transforms

Data cleaning, normalisation, and enrichment before write to S3 data lake.

Why PySpark? · Distributed processing · Handles TBs of data · 10-100× faster than single-node analytics · Native Python APIs

Scalable, Cost-Optimised Analytics Storage

RAW DATA LAYER

S3 / Bronze

All incoming events landed as-is in Parquet format. Immutable append-only storage. Full event replay capability.

- Watch event raw logs
- Search query streams
- API call traces
- Device telemetry

PROCESSED DATA LAYER

S3 / Silver

Cleansed, de-duplicated, and enriched data. Schema-enforced, partitioned by date and user segment.

- Session-aggregated metrics
- Churn feature vectors
- User engagement scores
- Content affinity maps

CURATED ANALYTICS LAYER

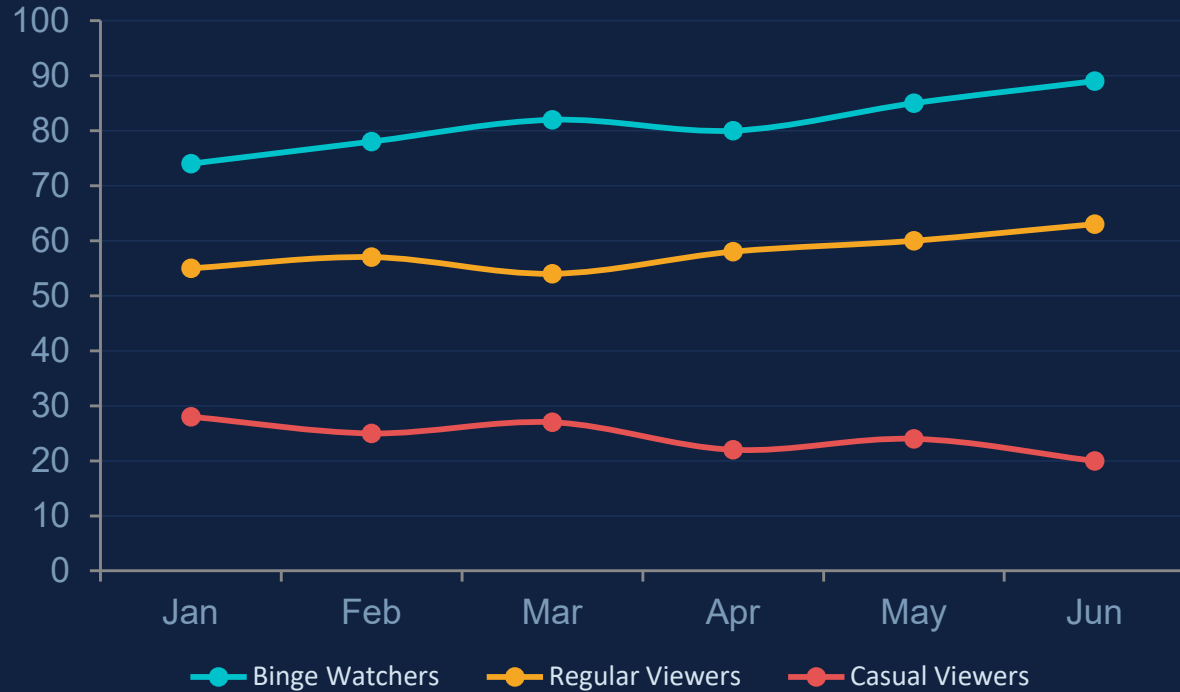
S3 / Gold + DWH

Analytics-ready data models powering Power BI dashboards and AI training pipelines. SLA-governed.

- CLV forecasts
- Churn risk scores
- Segment classifications
- Recommendation outputs

Decision Intelligence at a Glance

Monthly Engagement Score by Segment



Avg Watch Time

87 min

+12% MoM

Churn Risk Users

4,218

This Week

Avg CLV

\$312

Per Subscriber

Rec. CTR

34.2%

AI
Recommendations

Strategic Growth Levers

01

AI-Powered Segmentation

Continuously refresh user segments using real-time signals. Move beyond static cohorts to dynamic, intent-driven clusters.

↑ 18% retention

02

Personalised Retention Campaigns

Trigger automated retention journeys 14 days before predicted churn. A/B test incentives — content unlocks, price trials.

↓ 22% churn rate

03

Regional Content Optimisation

Identify title performance by geography. Commission region-specific content informed by local engagement analytics.

↑ 31% regional engagement

04

Dynamic Recommendation Strategy

Rotate recommendation algorithms based on subscriber lifecycle stage — exploration for new users, depth for established watchers.

↑ 27% watch time

05

Engagement-Based Loyalty System

Reward consistent viewing milestones with exclusive content access. Use CLV tier thresholds to prioritise high-value users.

↑ 15% LTV

Trust, Compliance & Responsible Data Use



GDPR Compliance

Lawful data processing, right to erasure, consent management, and DPA-mandated breach notifications within 72 hours.



Responsible AI Usage

Model explainability (SHAP values), fairness audits across demographic groups, and documented bias mitigation strategies.



CCPA Compliance

California residents' opt-out rights, data category disclosure, and no-sale guarantees for personal information.



Ethical Personalisation

No dark patterns — users can view, export and delete their data profile. Recommendation transparency via 'Why we suggested this'.



Data Anonymisation

PII stripped at ingestion via differential privacy and k-anonymity. Analytics operate on pseudonymised identifiers only.

Unlocking the World's Largest Streaming Market



Regional Language Personalisation

Deploy NLP pipelines for 12+ Asian languages. UI localisation, subtitle generation, and culturally-relevant genre taxonomies.



Localised Content Recommendations

Train market-specific recommendation models. Regional behavioural patterns differ significantly from Western cohorts.



Dynamic Pricing Models

Micro-tiered subscriptions calibrated to local purchasing power parity. Mobile-first plans for high-smartphone penetration markets.



Telecom Partnerships

Bundle CineWave with local carrier data plans. Zero-rating content drives acquisition in price-sensitive markets like India and SEA.

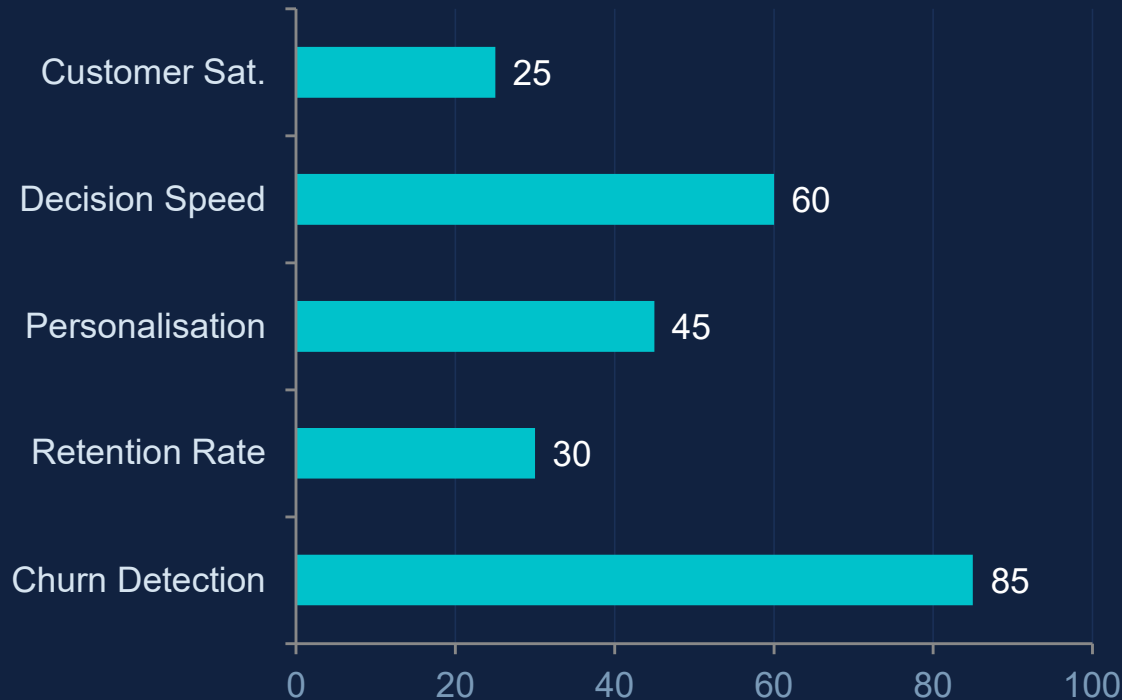


Behaviour-Driven Analytics

Market-specific KPI dashboards. Asian users exhibit distinct content discovery behaviours — search-led vs browse-led engagement.

Measurable Business & Technical Impact

Projected % Improvement Post-Implementation



Early Churn Detection

Predict cancellations 14 days in advance with 85%+ model accuracy.



Higher Retention

Targeted campaigns projected to reduce monthly churn by up to 30%.



Scalable Infrastructure

Cloud-native pipeline handles 10x subscriber growth without re-architecture.



Faster Decisions

Real-time dashboards cut reporting lag from days to seconds.

From OTT Platform to Intelligent Ecosystem

CineWave can transform from a traditional streaming platform into an intelligent, AI-driven, customer-centric ecosystem through scalable analytics architecture, real-time data processing, and ethical personalisation strategies.

Scalable Architecture

Real-Time Processing

AI-Driven Intelligence

Ethical Personalisation

The proposed solution improves subscriber engagement, reduces churn, enables operational intelligence, and positions CineWave for sustainable long-term growth in the evolving OTT landscape.

THANK YOU — Questions & Discussion Welcome